## NumPy Arrays & Operation

```
import numpy as np

arr1=np.array([10,20,30,40,50])
print("1D:",arr1)
arr2=np.array([[1,2,3,],[4,5,6],[7,8,9]])
print("\n2D:",arr2)
arr3=np.array([[[1,2],[3,4]],[[8,5],[6,7]]])
print("\n3D:",arr3)
```

```
1D: [10 20 30 40 50]

2D: [[1 2 3]
 [4 5 6]
 [7 8 9]]

3D: [[[1 2]
  [3 4]]

 [[8 5]
  [6 7]]]
```

## Arithmatic operation

```
a=np.array([10,20,30,40,50])
b=np.array([5,6,7,8,9])
print("Addition:",a+b)
print("\nSubtraction:",a-b)
print("\nMultiplication:",a*b)
print("\nDivision:",a/b)
```

```
Addition: [15 26 37 48 59]

Subtraction: [ 5 14 23 32 41]

Multiplication: [ 50 120 210 320 450]

Division: [2.         3.33333333 4.28571429 5.         5.55555556]
```

## NumPy Functions

```
arr=np.array([10,20,30,40,50,60])
mean=np.mean(arr)
print("mean:",mean)
median=np.median(arr)
print("median:",median)
std=np.std(arr)
print("std:",std)
sum=np.sum(arr)
print("sum:",sum)
reshape=np.reshape(arr,(2,3))
print("reshape:",reshape)
arange=np.arange(1,10)
print("arange:",arange)
linspace=np.linspace(1,10,5)
print("linspace:",linspace)

max=np.argmax(arr)
print("maximum index:",max)
min=np.argmin(arr)
print("minimum index:",min)
```

```
mean: 35.0
median: 35.0
std: 17.07825127659933
sum: 210
reshape: [[10 20 30]
 [40 50 60]]
arange: [1 2 3 4 5 6 7 8 9]
linspace: [ 1.    3.25  5.5   7.75 10.  ]
maximum index: 5
minimum index: 0
```

## Project Title

Penguins Data Analysis & Visualization Project

## Dataset Description

This dataset contains information about different penguin species including their physical measurements, habitat details, and biological attributes.

Key features include:

- Species
- Culmen Length & Depth
- Flipper Length
- Body Mass
- Sex
- Region & Island

The dataset is used to analyze patterns, relationships, and differences among penguin species.

## Import Libraries & Load Dataset

```
# Pandas Basics
import pandas as pd
url="https://raw.githubusercontent.com/gomathivvm-9496/penguin-data/5483aae33d27590712406c729d1871aa1a7fff00/penguins_lter.csv"

df=pd.read_csv(url)
df.head()
```

| | studyName | Sample Number | Species | Region | Island | Stage | Individual ID | Clutch Completion | Date Egg | Culmen Length (mm) | Culmen Depth (mm) | Flipper Length (mm) | Body Mass (g) | S |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | PAL0708 | 1 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N1A1 | Yes | 11/11/07 | 39.1 | 18.7 | 181.0 | 3750.0 | MAL |
| **1** | PAL0708 | 2 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N1A2 | Yes | 11/11/07 | 39.5 | 17.4 | 186.0 | 3800.0 | FEMAL |
| **2** | PAL0708 | 3 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N2A1 | Yes | 11/16/07 | 40.3 | 18.0 | 195.0 | 3250.0 | FEMAL |
| **3** | PAL0708 | 4 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N2A2 | Yes | 11/16/07 | NaN | NaN | NaN | NaN | Na |
| **4** | PAL0708 | 5 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N3A1 | Yes | 11/16/07 | 36.7 | 19.3 | 193.0 | 3450.0 | FEMAL |

```
df.tail()
```

| | studyName | Sample Number | Species | Region | Island | Stage | Individual ID | Clutch Completion | Date Egg | Culmen Length (mm) | Culmen Depth (mm) | Flipper Length (mm) | Body Mass (g) | S |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **339** | PAL0910 | 120 | Gentoo penguin (Pygoscelis papua) | Anvers | Biscoe | Adult, 1 Egg Stage | N38A2 | No | 12/1/09 | NaN | NaN | NaN | NaN | Na |
| **340** | PAL0910 | 121 | Gentoo penguin (Pygoscelis papua) | Anvers | Biscoe | Adult, 1 Egg Stage | N39A1 | Yes | 11/22/09 | 46.8 | 14.3 | 215.0 | 4850.0 | FEMAL |
| **341** | PAL0910 | 122 | Gentoo penguin (Pygoscelis papua) | Anvers | Biscoe | Adult, 1 Egg Stage | N39A2 | Yes | 11/22/09 | 50.4 | 15.7 | 222.0 | 5750.0 | MAL |
| **342** | PAL0910 | 123 | Gentoo penguin (Pygoscelis papua) | Anvers | Biscoe | Adult, 1 Egg Stage | N43A1 | Yes | 11/22/09 | 45.2 | 14.8 | 212.0 | 5200.0 | FEMAL |
| **343** | PAL0910 | 124 | Gentoo penguin (Pygoscelis papua) | Anvers | Biscoe | Adult, 1 Egg Stage | N43A2 | Yes | 11/22/09 | 49.9 | 16.1 | 213.0 | 5400.0 | MAL |

```
df.columns
```

```
Index(['studyName', 'Sample Number', 'Species', 'Region', 'Island', 'Stage',
       'Individual ID', 'Clutch Completion', 'Date Egg', 'Culmen Length (mm)',
       'Culmen Depth (mm)', 'Flipper Length (mm)', 'Body Mass (g)', 'Sex',
       'Delta 15 N (o/oo)', 'Delta 13 C (o/oo)', 'Comments'],
      dtype='object')
```

```
df.shape
```

```
(344, 17)
```

```
df.dtypes
```

| | 0 |
|---|---|
| **studyName** | object |
| **Sample Number** | int64 |
| **Species** | object |
| **Region** | object |
| **Island** | object |
| **Stage** | object |
| **Individual ID** | object |
| **Clutch Completion** | object |
| **Date Egg** | object |
| **Culmen Length (mm)** | float64 |
| **Culmen Depth (mm)** | float64 |
| **Flipper Length (mm)** | float64 |
| **Body Mass (g)** | float64 |
| **Sex** | object |
| **Delta 15 N (o/oo)** | float64 |
| **Delta 13 C (o/oo)** | float64 |
| **Comments** | object |

**dtype:** object

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 17 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   studyName            344 non-null    object
 1   Sample Number        344 non-null    int64
 2   Species              344 non-null    object
 3   Region               344 non-null    object
 4   Island               344 non-null    object
 5   Stage                344 non-null    object
 6   Individual ID        344 non-null    object
 7   Clutch Completion    344 non-null    object
 8   Date Egg             344 non-null    object
 9   Culmen Length (mm)   342 non-null    float64
 10  Culmen Depth (mm)    342 non-null    float64
 11  Flipper Length (mm)  342 non-null    float64
 12  Body Mass (g)        342 non-null    float64
 13  Sex                  334 non-null    object
 14  Delta 15 N (o/oo)    330 non-null    float64
 15  Delta 13 C (o/oo)    331 non-null    float64
 16  Comments             26 non-null     object
dtypes: float64(6), int64(1), object(10)
memory usage: 45.8+ KB
```

```
df["Sex"].unique()
```

```
array(['MALE', 'FEMALE', nan, '.'], dtype=object)
```

```
df.describe()
```

| | Sample Number | Culmen Length (mm) | Culmen Depth (mm) | Flipper Length (mm) | Body Mass (g) | Delta 15 N (o/oo) | Delta 13 C (o/oo) |
|---|---|---|---|---|---|---|---|
| **count** | 344.000000 | 342.000000 | 342.000000 | 342.000000 | 342.000000 | 330.000000 | 331.000000 |
| **mean** | 63.151163 | 43.921930 | 17.151170 | 200.915205 | 4201.754386 | 8.733382 | -25.686292 |
| **std** | 40.430199 | 5.459584 | 1.974793 | 14.061714 | 801.954536 | 0.551770 | 0.793961 |
| **min** | 1.000000 | 32.100000 | 13.100000 | 172.000000 | 2700.000000 | 7.632200 | -27.018540 |
| **25%** | 29.000000 | 39.225000 | 15.600000 | 190.000000 | 3550.000000 | 8.299890 | -26.320305 |
| **50%** | 58.000000 | 44.450000 | 17.300000 | 197.000000 | 4050.000000 | 8.652405 | -25.833520 |
| **75%** | 95.250000 | 48.500000 | 18.700000 | 213.000000 | 4750.000000 | 9.172123 | -25.062050 |

## Data Cleaning Steps

```
df.columns = df.columns.str.strip()
df.columns = df.columns.str.replace(" ", "_")
df.columns = df.columns.str.replace("(", "").str.replace(")", "")
```

```
#Add Columns
df["Large_penguin"]=df["Body_Mass_g"]>4000
df.head()
```

| | studyName | Sample_Number | Species | Region | Island | Stage | Individual_ID | Clutch_Completion | Date_Egg | Culmen_Length_mm | Cu |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | PAL0708 | 1 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N1A1 | Yes | 11/11/07 | 39.1 | |
| **1** | PAL0708 | 2 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N1A2 | Yes | 11/11/07 | 39.5 | |
| **2** | PAL0708 | 3 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N2A1 | Yes | 11/16/07 | 40.3 | |
| **3** | PAL0708 | 4 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N2A2 | Yes | 11/16/07 | NaN | |
| **4** | PAL0708 | 5 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N3A1 | Yes | 11/16/07 | 36.7 | |

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 18 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   studyName          344 non-null    object
 1   Sample_Number      344 non-null    int64
 2   Species            344 non-null    object
 3   Region             344 non-null    object
 4   Island             344 non-null    object
 5   Stage              344 non-null    object
 6   Individual_ID      344 non-null    object
 7   Clutch_Completion  344 non-null    object
 8   Date_Egg           344 non-null    object
 9   Culmen_Length_mm   342 non-null    float64
 10  Culmen_Depth_mm    342 non-null    float64
 11  Flipper_Length_mm  342 non-null    float64
 12  Body_Mass_g        342 non-null    float64
 13  Sex                334 non-null    object
 14  Delta_15_N_o/oo    330 non-null    float64
 15  Delta_13_C_o/oo    331 non-null    float64
 16  Comments           26 non-null     object
 17  Large_penguin      344 non-null    bool
dtypes: bool(1), float64(6), int64(1), object(10)
memory usage: 46.2+ KB
```

```
#update column
df["Culmen_Length_mm"]=df["Culmen_Length_mm"].fillna(df["Culmen_Length_mm"].mean())
df["Culmen_Depth_mm"]=df["Culmen_Depth_mm"].fillna(df["Culmen_Depth_mm"].mean())
df["Flipper_Length_mm"]=df["Flipper_Length_mm"].fillna(df["Flipper_Length_mm"].mean())
df["Body_Mass_g"]=df["Body_Mass_g"].fillna(df["Body_Mass_g"].mean())
```

```
#Delete Columns
df.drop("Comments",axis=1,inplace=True)
df.head()
```

| | studyName | Sample_Number | Species | Region | Island | Stage | Individual_ID | Clutch_Completion | Date_Egg | Culmen_Length_mm | Cu |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | PAL0708 | 1 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N1A1 | Yes | 11/11/07 | 39.10000 | |
| **1** | PAL0708 | 2 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N1A2 | Yes | 11/11/07 | 39.50000 | |
| **2** | PAL0708 | 3 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N2A1 | Yes | 11/16/07 | 40.30000 | |
| **3** | PAL0708 | 4 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N2A2 | Yes | 11/16/07 | 43.92193 | |
| **4** | PAL0708 | 5 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N3A1 | Yes | 11/16/07 | 36.70000 | |

```
df.set_index("Sample_Number",inplace=True)
df.head()
```

| Sample_Number | studyName | Species | Region | Island | Stage | Individual_ID | Clutch_Completion | Date_Egg | Culmen_Length_mm | Culm |
|---|---|---|---|---|---|---|---|---|---|---|
| **1** | PAL0708 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N1A1 | Yes | 11/11/07 | 39.10000 | |
| **2** | PAL0708 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N1A2 | Yes | 11/11/07 | 39.50000 | |
| **3** | PAL0708 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N2A1 | Yes | 11/16/07 | 40.30000 | |
| **4** | PAL0708 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N2A2 | Yes | 11/16/07 | 43.92193 | |
| **5** | PAL0708 | Adelie Penguin (Pygoscelis adeliae) | Anvers | Torgersen | Adult, 1 Egg Stage | N3A1 | Yes | 11/16/07 | 36.70000 | |

```
df.sort_values("Body_Mass_g",ascending=False).head()
```

| Sample_Number | studyName | Species | Region | Island | Stage | Individual_ID | Clutch_Completion | Date_Egg | Culmen_Length_mm | Culmen_ |
|---|---|---|---|---|---|---|---|---|---|---|
| **18** | PAL0708 | Gentoo penguin (Pygoscelis papua) | Anvers | Biscoe | Adult, 1 Egg Stage | N39A2 | Yes | 11/27/07 | 49.2 | |
| **34** | PAL0708 | Gentoo penguin (Pygoscelis papua) | Anvers | Biscoe | Adult, 1 Egg Stage | N56A2 | Yes | 12/3/07 | 59.6 | |
| **118** | PAL0910 | Gentoo penguin (Pygoscelis papua) | Anvers | Biscoe | Adult, 1 Egg Stage | N36A2 | Yes | 12/1/09 | 48.8 | |
| **78** | PAL0809 | Gentoo penguin (Pygoscelis papua) | Anvers | Biscoe | Adult, 1 Egg Stage | N58A2 | Yes | 11/6/08 | 51.1 | |
| **80** | PAL0809 | Gentoo penguin (Pygoscelis papua) | Anvers | Biscoe | Adult, 1 Egg Stage | N60A2 | Yes | 11/9/08 | 45.2 | |

## Data Exploration & Wrangling

```
df["Species"]=df["Species"].str.split("(").str[0].str.strip()
df["Species"]
```

| Sample_Number | Species |
|---|---|
| **1** | Adelie Penguin |
| **2** | Adelie Penguin |
| **3** | Adelie Penguin |
| **4** | Adelie Penguin |
| **5** | Adelie Penguin |
| **...** | ... |
| **120** | Gentoo penguin |
| **121** | Gentoo penguin |
| **122** | Gentoo penguin |
| **123** | Gentoo penguin |
| **124** | Gentoo penguin |

344 rows × 1 columns

**dtype:** object

```
df["Species"].value_counts()
```

| Species | count |
|---|---|
| **Adelie Penguin** | 152 |
| **Gentoo penguin** | 124 |
| **Chinstrap penguin** | 68 |

**dtype:** int64

```
import numpy as np
import pandas as pd
df["Sex"]=df["Sex"].replace(".",np.nan)
```

```
df["Sex"]=df["Sex"].fillna(df["Sex"].mode()[0])
df['Date_Egg'] = pd.to_datetime(df['Date_Egg'], format='%m/%d/%y')
df['year'] = df['Date_Egg'].dt.year
```

```
cols=["Delta_15_N_o/oo","Delta_13_C_o/oo"]
for col in cols:
  df[col]=df[col].fillna(df[col].mean())
df.head()
```

| Sample_Number | studyName | Species | Region | Island | Stage | Individual_ID | Clutch_Completion | Date_Egg | Culmen_Length_mm | Culmen_ |
|---|---|---|---|---|---|---|---|---|---|---|
| **1** | PAL0708 | Adelie Penguin | Anvers | Torgersen | Adult, 1 Egg Stage | N1A1 | Yes | 2007-11-11 | 39.10000 | |
| **2** | PAL0708 | Adelie Penguin | Anvers | Torgersen | Adult, 1 Egg Stage | N1A2 | Yes | 2007-11-11 | 39.50000 | |
| **3** | PAL0708 | Adelie Penguin | Anvers | Torgersen | Adult, 1 Egg Stage | N2A1 | Yes | 2007-11-16 | 40.30000 | |
| **4** | PAL0708 | Adelie Penguin | Anvers | Torgersen | Adult, 1 Egg Stage | N2A2 | Yes | 2007-11-16 | 43.92193 | |
| **5** | PAL0708 | Adelie Penguin | Anvers | Torgersen | Adult, 1 Egg Stage | N3A1 | Yes | 2007-11-16 | 36.70000 | |

```
df.isnull().sum()
```

| | 0 |
|---|---|
| **studyName** | 0 |
| **Species** | 0 |
| **Region** | 0 |
| **Island** | 0 |
| **Stage** | 0 |
| **Individual ID** | 0 |
| **Clutch Completion** | 0 |
| **Date Egg** | 0 |
| **Culmen Length (mm)** | 0 |
| **Culmen Depth (mm)** | 0 |
| **Flipper Length (mm)** | 0 |
| **Body Mass (g)** | 0 |
| **Sex** | 0 |
| **Delta 15 N (o/oo)** | 0 |
| **Delta 13 C (o/oo)** | 0 |
| **Large_penguin** | 0 |

**dtype:** int64

```
df["Body_Mass_g"].fillna(df["Body_Mass_g"].mean(), inplace=True)
```

```
/tmp/ipykernel_14831/2080037970.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chai
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are sett

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df
```

```
df["Body_Mass_g"].fillna(df["Body_Mass_g"].mean(), inplace=True)
```

```
for col in df.columns:
    print(repr(col))
```

```
'studyName'
'Sample_Number'
'Species'
'Region'
'Island'
'Stage'
'Individual_ID'
'Clutch_Completion'
'Date_Egg'
'Culmen_Length_mm'
'Culmen_Depth_mm'
'Flipper_Length_mm'
'Body_Mass_g'
'Sex'
'Delta_15_N_o/oo'
'Delta_13_C_o/oo'
'Comments'
```

## Feature Engineering

```
df["size"]=df["Body_Mass_g"].apply(lambda x:"Small" if x<3500 else("Medium" if x<4500 else "Large"))
```

```
#Group By and Aggregate Data
df.groupby("Species")["Body_Mass_g"].agg(["mean","max","min"])
```

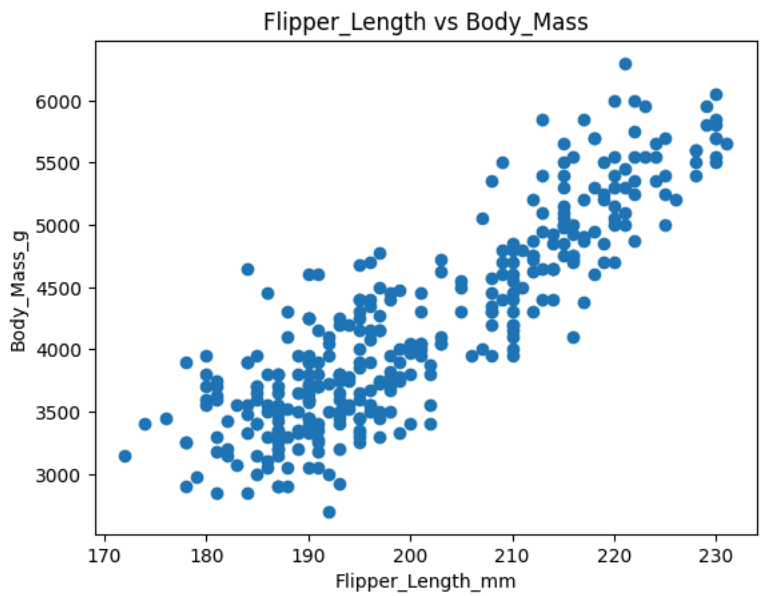
| Species | mean | max | min |
|---|---|---|---|
| **Adelie Penguin** | 3700.662252 | 4775.0 | 2850.0 |
| **Chinstrap penguin** | 3733.088235 | 4800.0 | 2700.0 |
| **Gentoo penguin** | 5076.016260 | 6300.0 | 3950.0 |

```
df.reset_index(inplace=True)
```

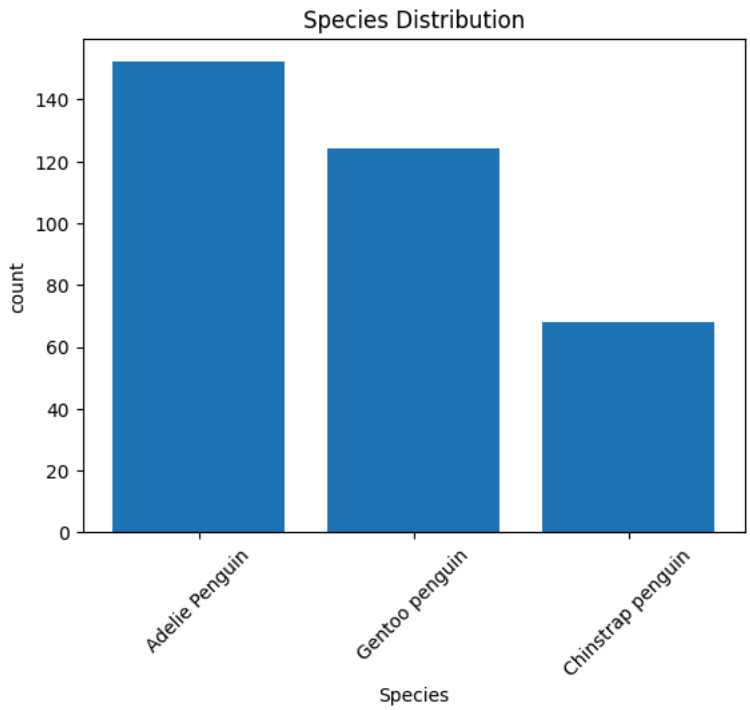
## Visualizations

```
#Matplotlib Visualization
#Line chart
import matplotlib.pyplot as plt
plt.plot(df["Sample_Number"],df["Body_Mass_g"])
plt.xlabel("Sample_Number")
plt.ylabel("Body_Mass_g")
plt.title("Body Mass trend")
plt.show()
```

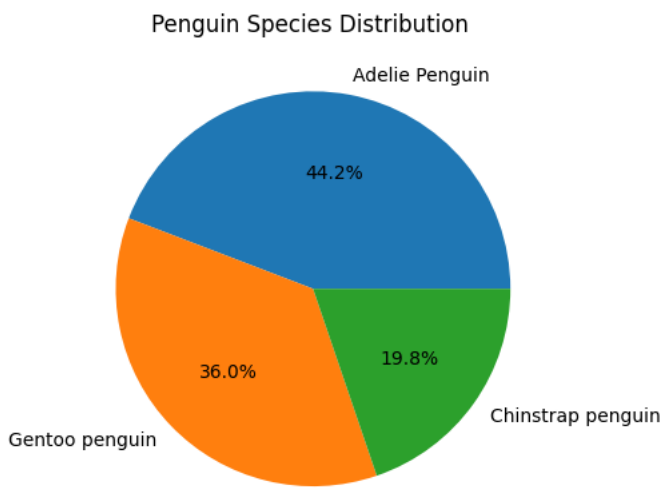
```
#Scatter Plot
plt.scatter(df["Flipper_Length_mm"],df["Body_Mass_g"])
plt.xlabel("Flipper_Length_mm")
plt.ylabel("Body_Mass_g")
plt.title("Flipper_Length vs Body_Mass")
plt.show()
```



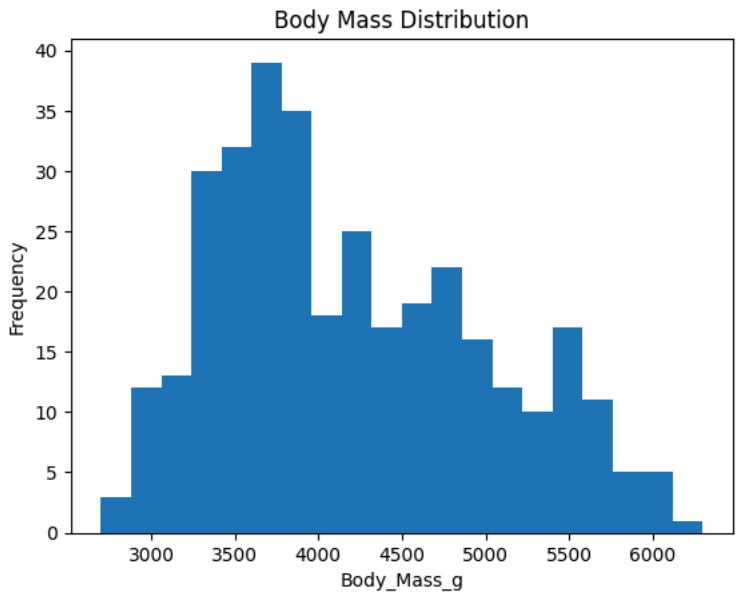

```
#Bar Chart
species_counts=df["Species"].value_counts()
plt.bar(species_counts.index,species_counts.values)
plt.xlabel("Species")
plt.ylabel("count")
plt.title("Species Distribution")
plt.xticks(rotation=45)
plt.show()
```



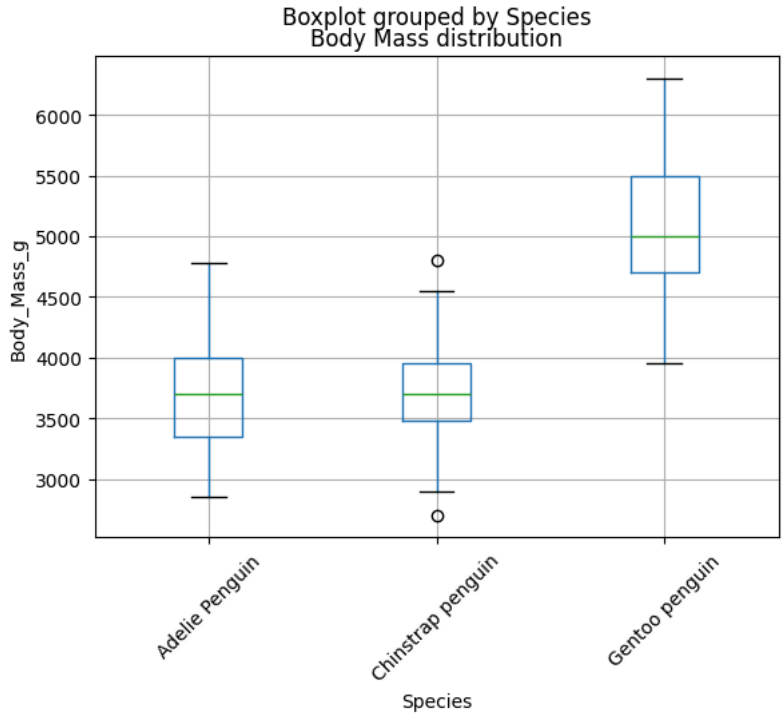

```
#Pie Chart
Species_count=df["Species"].value_counts()
plt.pie(Species_count.values,labels=Species_count.index,autopct="%1.1f%%")
plt.title("Penguin Species Distribution ")
plt.show()
```



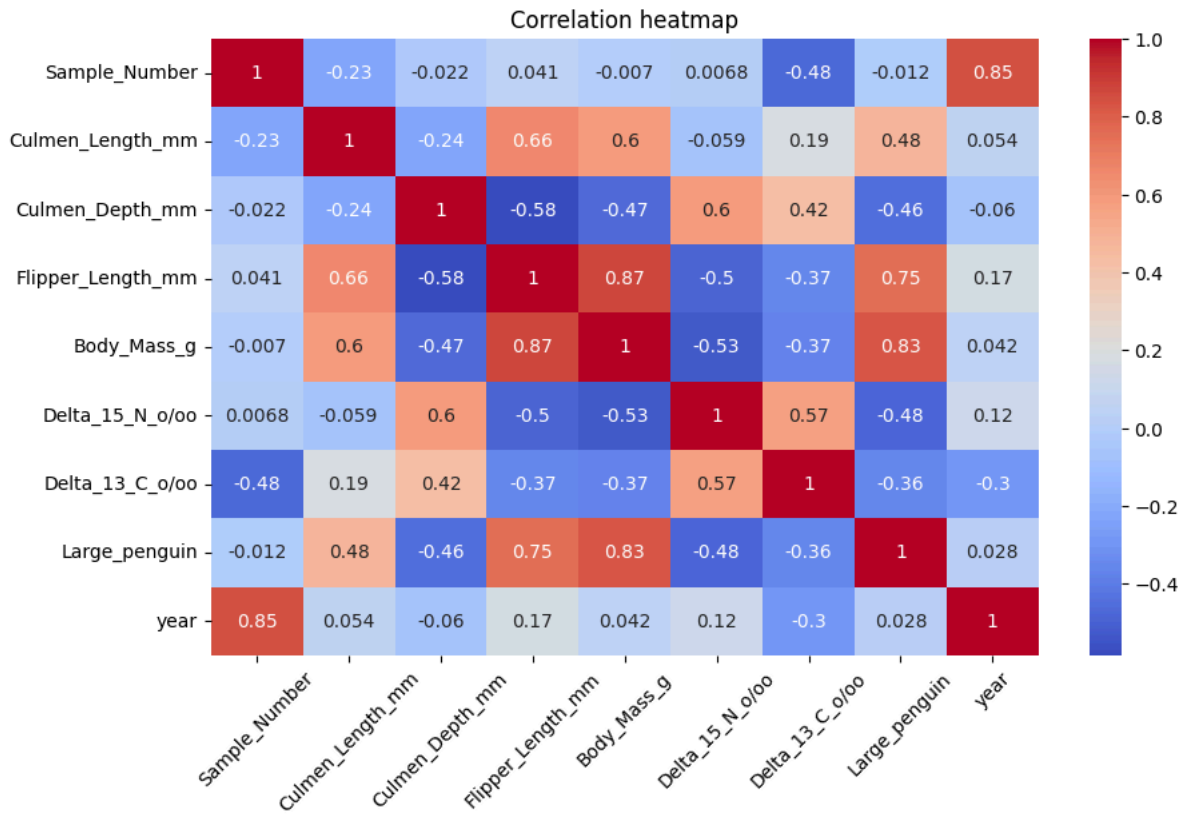

```
#Histogram
plt.hist(df["Body_Mass_g"],bins=20)
plt.xlabel("Body_Mass_g")
plt.ylabel("Frequency")
plt.title("Body Mass Distribution")
plt.show()
```



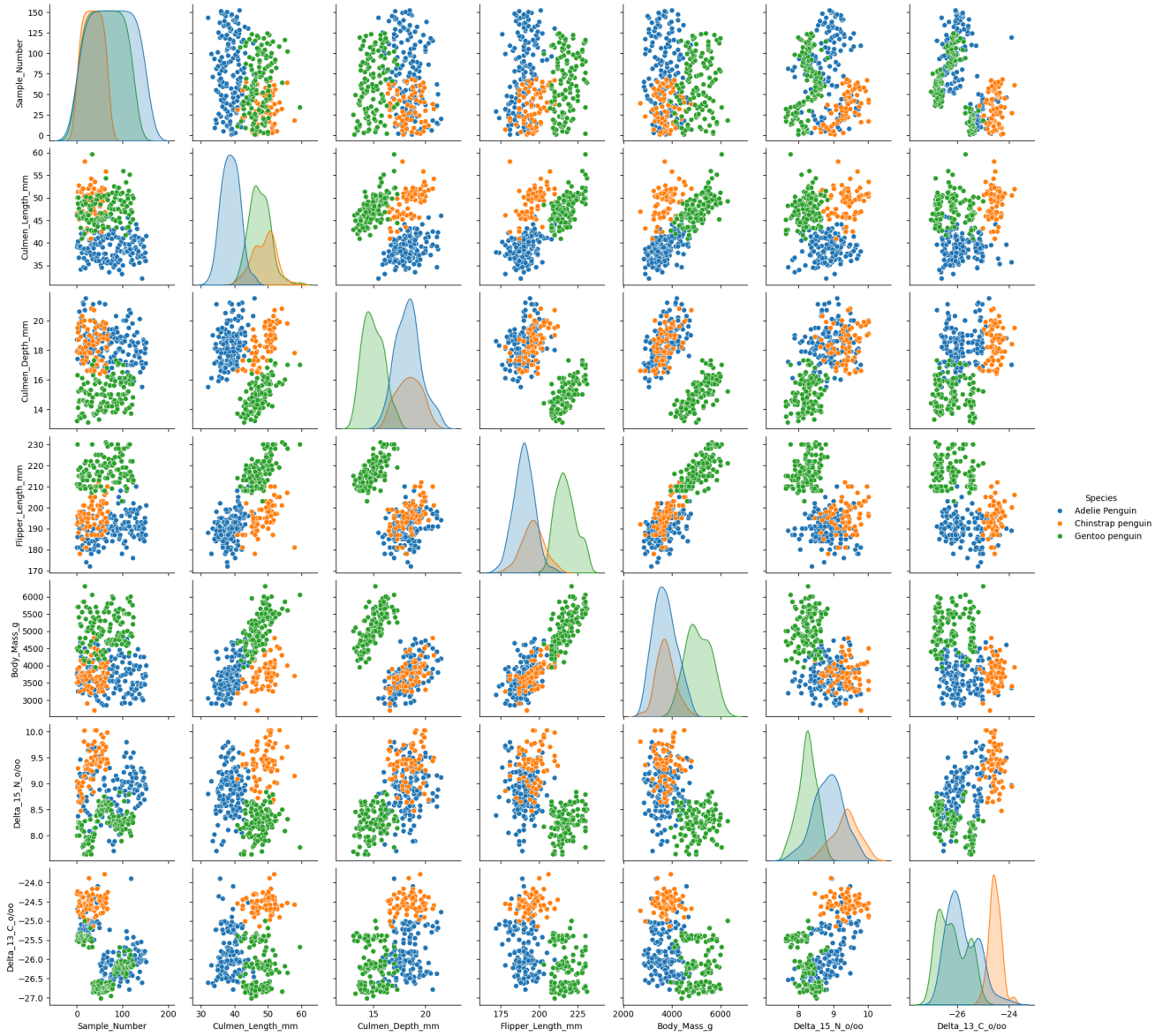

```
#Box Plot
df.boxplot(column="Body_Mass_g",by="Species")
plt.xlabel("Species")
plt.ylabel("Body_Mass_g")
plt.title("Body Mass distribution")
plt.xticks(rotation=45)
plt.show()
```



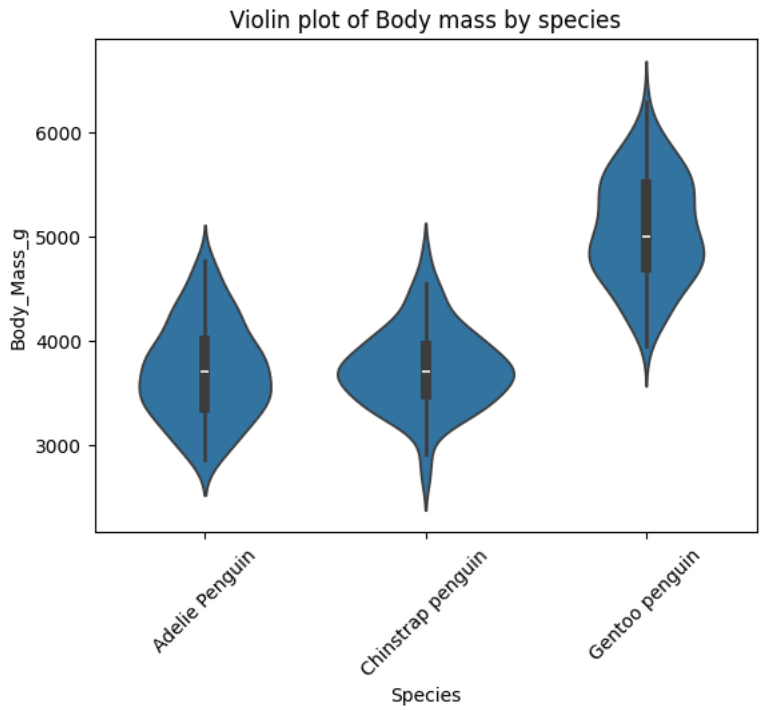

```
#Seaborn & Plotly Advanced Visualization
#Heatmap
import seaborn as sns
corr=df.corr(numeric_only=True)
plt.figure(figsize=(10,6))
sns.heatmap(corr,annot=True,cmap="coolwarm")
plt.title("Correlation heatmap")
plt.xticks(rotation=45)
plt.show()
```



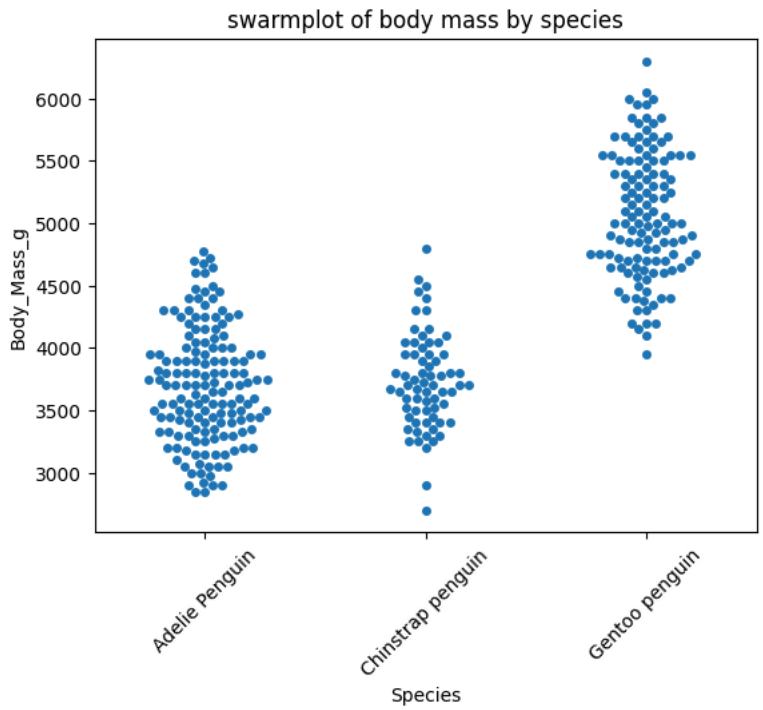

```
#pair plots
sns.pairplot(df,hue="Species")
plt.show()
```



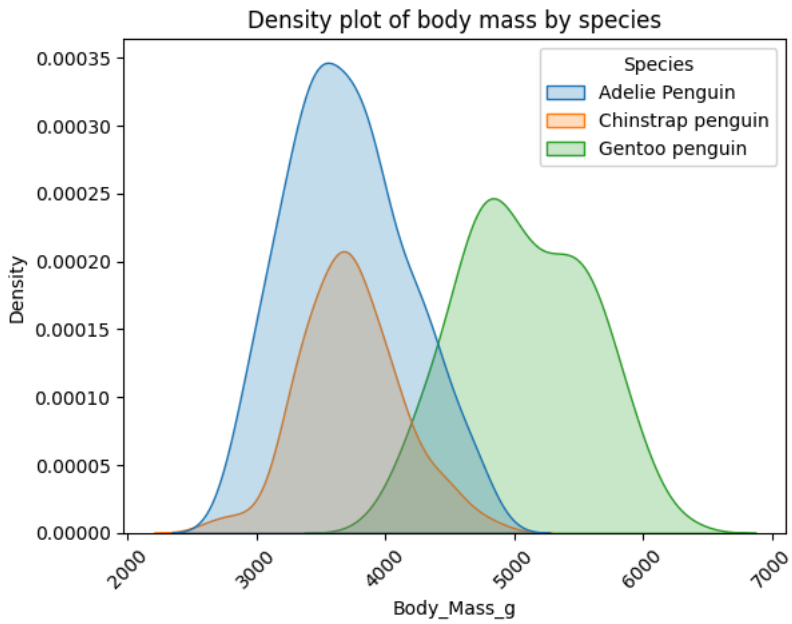

```
 #violin
sns.violinplot(x="Species",y="Body_Mass_g",data=df)
plt.title("Violin plot of Body mass by species")
plt.xticks(rotation=45)
plt.show()
```



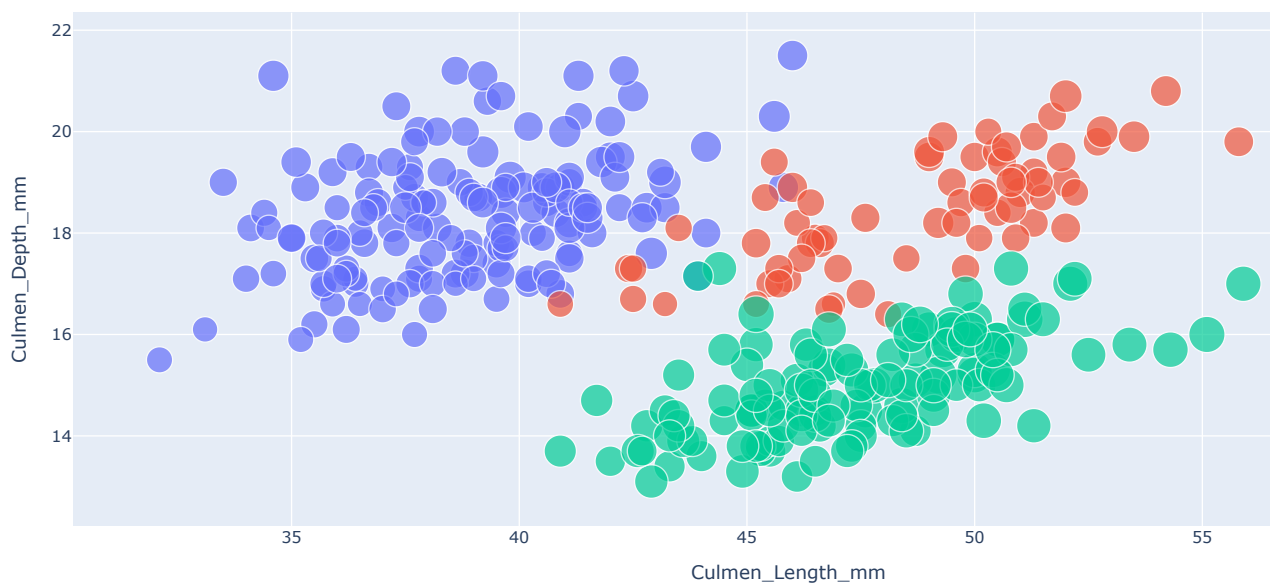

```
#swarm
sns.swarmplot(x="Species",y="Body_Mass_g",data=df)
plt.title("swarmplot of body mass by species")
plt.xticks(rotation=45)
plt.show()
```




```
#density plots
sns.kdeplot(df,x="Body_Mass_g",hue="Species",fill=True)
plt.title("Density plot of body mass by species")
plt.xticks(rotation=45)
plt.show()
```



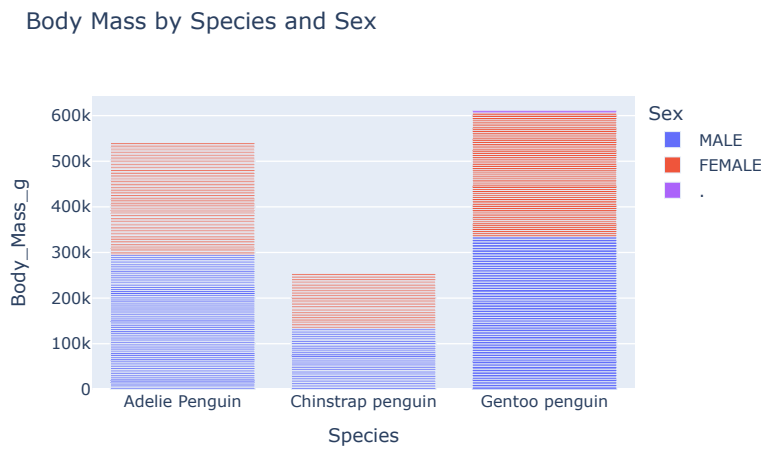

```
#Plotly Interactive Visualizations
#scatter
import plotly.express as px
fig=px.scatter(df,x="Culmen_Length_mm",y="Culmen_Depth_mm",color="Species",size="Body_Mass_g")
fig.show()
```



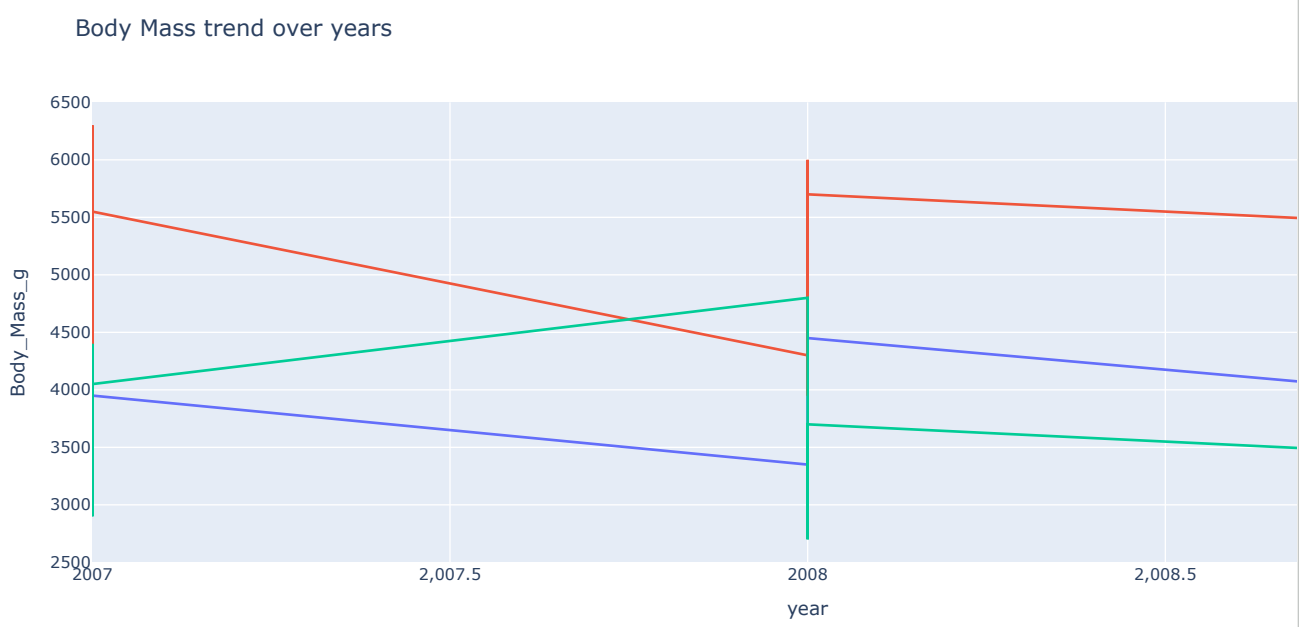

```
df["Sex"].unique()
```

```
array(['MALE', 'FEMALE', '.'], dtype=object)
```

```
#bar
fig=px.bar(df,x="Species",y="Body_Mass_g",color="Sex",title="Body Mass by Species and Sex",width=600,height=400)
fig.show()
```




```
#line
df_sorted=df.sort_values(by="year")
fig=px.line(df_sorted,x="year",y="Body_Mass_g",color="Species",title="Body Mass trend over years")
fig.show()
```




## Data Cleaning & Stats

```
!pip install requests
```

```
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (2.32.4)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests) (2026.2.25)
```

```
print(response.status_code)
```

```
200
```

## Web Scraping Integration

```
import requests
from bs4 import BeautifulSoup

url = "https://en.wikipedia.org/wiki/Penguin"

headers = {"User-Agent": "Mozilla/5.0"}

response = requests.get(url, headers=headers)

html_content = response.text

soup = BeautifulSoup(html_content, "html.parser")

if soup.title:
    print(soup.title.text)
else:
    print("Title not found")
```

```
Penguin - Wikipedia
```

```
print(html_content[:500])
```

```
<!DOCTYPE html>
<html class="client-nojs vector-feature-language-in-header-enabled vector-feature-language-in-main-menu-disabled vector-feature-
```

## AI Assited Analysis

```
# Find all heading tags
headings = soup.find_all(['h1', 'h2', 'h3', 'h4', 'h5', 'h6'])

# Extract and print the text from each heading
for heading in headings:
    print(f"{heading.name}: {heading.text.strip()}")
    # Instead of pd.read_html(url), use the content you already downloaded:
tables = pd.read_html(html_content)

# Now you can continue with your merge

species_info = pd.DataFrame({
    'Species': ['Adelie Penguin', 'Gentoo Penguin', 'Chinstrap Penguin'],
    'Conservation_Status': ['Least Concern', 'Least Concern', 'Least Concern'],
    'Average_Height_cm': [70, 80, 72]
})
df_final = pd.merge(df, species_info, on='Species', how='left')
df_final.head()
```

```
h2: Contents
h1: Penguin
h2: Etymology
h3: Pinguinus
h2: Systematics and evolution
h3: Taxonomy
h3: Evolution
h4: Basal fossils
h4: Palaeeudyptines
h4: Origin and systematics of modern penguins
h4: Geography
h4: Relationship to other bird orders
h2: Anatomy and physiology
h2: Behaviour
h3: Breeding
h2: Distribution and habitat
h3: Conservation status
h2: Penguins and humans
h3: In popular culture
h2: References
h2: Bibliography
h2: External links
/tmp/ipykernel_14831/2729916342.py:8: FutureWarning:
```

```
#Extract Useful Data
from bs4 import BeautifulSoup
soup=BeautifulSoup(html_content,"html.parser")
paragraphs=soup.find_all("p")
data=[p.text for p in paragraphs[:5]]
print(data)
```

```
['\n', 'For fossil taxa, see the list of penguins (fossil species)\n', 'Penguins are a group of flightless semi-aquatic sea bird
```

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Stage | | | | | | |
| 1 | 2 | PAL0708 | Adelie Penguin | Anvers | Torgersen | Adult, 1 Egg Stage | N1A2 | Yes | 2007-11-11 | | | 39.50000 |

```
#scraped data frame
```